

By a graph we mean a graph in which each of ①-④ holds:

- ① there is 1 vertex colored red & all other vertices are colored purple
- ② there is 1 edge colored red & all other vertices are colored either black or purple
- ③ those colored either purple or red are called "colored edges"
- ④ multi-edges are allowed, but for each pair of vertices, there's at most 1 colored edge connecting that pair of vertices
- ⑤ loops are allowed, but only black edges can be loops

Define

$$L = \{x_1, \bar{x}_1, \dots, x_5, \bar{x}_5\}$$

$$\bar{x}_j = x_j$$

$x_1, \bar{x}_1, \dots, x_5, \bar{x}_5$ is the increasing order on L

A distinguished admissible ltt structure is a connected graph w/

- 5 pairs of vertices labeled $x_1, \bar{x}_1, \dots, x_5, \bar{x}_5$
- x_1 is colored red, the remaining vertices are colored purple
- x_1 is connected by a red edge to x_2
- $\{\bar{x}_1, x_2, \bar{x}_2, \dots, x_5, \bar{x}_5\}$ is partitioned into 3 sets of 3 st the vertices within each partitioned are connected by purple edges
- each pair of vertices $\{x_j, \bar{x}_j\}$ are connected by black edges

2 colored graphs G_1, G_2 w/ vertices labeled by L are ep-isomorphic if $\exists$ a bijection φ of L st ①-② hold:

① for each $x \in L$, the vertices labeled x & $\varphi(x)$ have the same color

② for each $x, y \in L$ w/ $x \neq y$,

the vertices labeled by $\varphi(x)$ & $\varphi(y)$ are connected by an edge of color C iff the vertices labeled by x & y are connected by an edge of color C

An admissible ltt structure is any graph ep-isomorphic to a distinguished admissible ltt structure

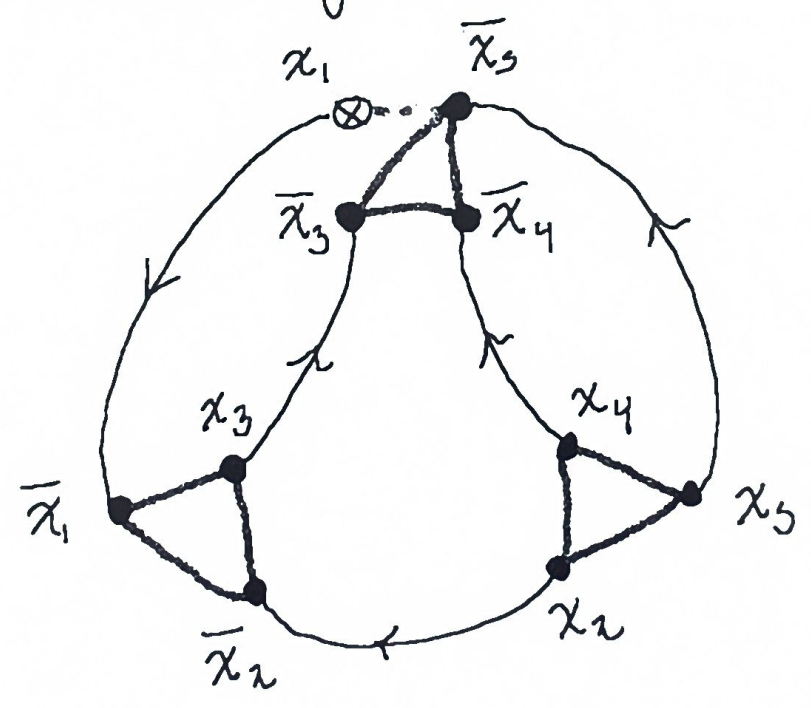
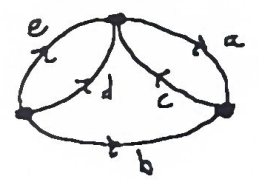
* ⁵⁶ have a code written to list all admissible ltt structures*
(these will give potential nodes of the automaton)

Place some (any) order on the admissible ltt structures.
 Let $\boxed{PLTT}$ denote this ordered list



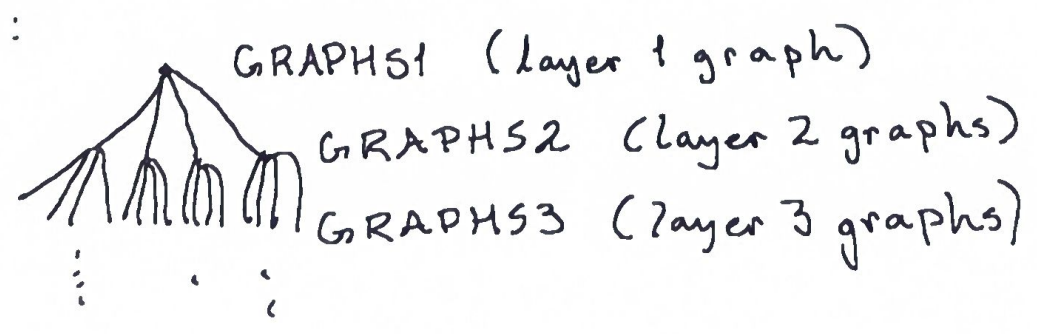
• Step 1: Start w/ a particular ltt structure $\mathbb{D}$

- $GRAPHS = \{\mathbb{D}\}$
- $graphs_1 = \{\mathbb{D}\}$
- $boxes = \{graphs_1\}$
- say that layer 1 consists of just $\mathbb{D}$, i.e. $GRAPHS_1 = \mathbb{D}$



R_k : $GRAPHS$ will ultimately be the list of all nodes in the automaton

- each $graphs_k$ will be a "box" of ep-isomorphic nodes
- boxes will be the collection of these ep-isomorphism classes
- Before closing up loops when nodes represent the same ep-isomorphism class of ltt structures, the automaton looks something like:



Recursively (wrt k) complete in PLTT order Steps 2-3 for 3
 each $\mathcal{G} \in \text{GRAPHSK}$:

Step 2: Find the 4 arrows: terminal Ltt structures out of $\mathcal{G}$:

Let k denote the layer of $\mathcal{G}$, i.e. $\mathcal{G} \in \text{GRAPHSK}$.

- Input the starting LTT structure $\mathcal{G}$ (will be an admissible ltt structure)
 - Vertices labeled w/ $\mathcal{L}$ • Vertex colors • Edges: edge colors
 - Call the label of the red vertex d_r
 - Call the label of the purple vertex of the red edge d_p
 - Call the label of the purple vertex of the red edge d_p
 - There are 2 vertices connected to the d_p vertex by purple edges:
 - ① Let d_1 be the label of the 1st of these } wrt the order on $\mathcal{L}$
 - ② Let d_2 be the label of the 2nd of these }
- For each of ① & ② there are 2 potential edges directed from $\mathcal{G}$ (i.e. then a potential terminal ltt structure for each of those edges):

For $i \in \{1, 2\}$:

- ⓐ the edge is labeled $d_r \mapsto d_i d_r$
 the terminal ltt structure $\mathcal{G}_{ia}$ has
 - red vertex d_r & red edge $\{d_r, \bar{d}_i\}$
 - precisely the following purple edges:
 - a purple edge $\{d_i, d\}$ whenever $\mathcal{G}$ has a colored edge $\{d_r, d\}$
 - a purple edge $\{d, d'\}$ whenever $\mathcal{G}$ has a purple edge $\{d, d'\}$
 - + neither $d = d_i$ nor $d' = d_i$

- ⓑ the edge is labeled $d_i \mapsto d_r d_i$
 the terminal ltt structure $\mathcal{G}_{ib}$ has
 - red vertex d_i & red edge $\{d_i, \bar{d}_r\}$
 - precisely the following purple edges
 - a purple edge $\{d_r, d\}$ whenever $\mathcal{G}$ has a colored edge $\{d_i, d\}$
 - a purple edge $\{d, d'\}$ whenever $\mathcal{G}$ has a purple edge $\{d, d'\}$
 - + neither $d = d_r$ nor $d' = d_r$

- Let $\text{terminal-}\mathcal{G} := \{\mathcal{G}_{ia}, \mathcal{G}_{ib}, \mathcal{G}_{2a}, \mathcal{G}_{2b}\}$
 (repetitions allowed).

- Remove any graph from $\text{terminal-}\mathcal{G}$ that is not in PLTT

- Each graph of $\text{terminal-}\mathcal{G}$ is in level $k+1$

- $\text{terminal-}\mathcal{G}$ & level $k+1$ both inherit their order from PLTT

Step 3:

In PLTT order, the following is applied to each $D' \in \text{terminal} - D$

• If D' is epi-isomorphic to some $D'' \in \text{GRAPHS}$, but isn't D'' :

- $\text{GRAPHS} = \text{GRAPHS} + \{D'\}$
- $\text{graphs}_j = \text{graphs}_j + \{D'\}$, where $D'' \in \text{graph}_j$
- add a "permutation arrow" $D' \rightarrow D''$, labeled w/ φ , where φ is the permutation of L from the ep-isomorphism defn

• If there is some $D'' \in \text{GRAPHS}$ st $D' = D''$:
add an arrow $D' \rightarrow D''$ w/ the map $D' \rightarrow D''$

• For the remaining D' ,

• $\text{GRAPHS} = \text{GRAPHS} + \{D'\}$

• $\text{graphs}_{j+1} = \{D'\} \cup \text{graphs}_j \in \text{boxes } i, \text{graphs}_{j+1} \notin \text{boxes}$

• $\text{GRAPHS}(k+1) = \text{GRAPHS}(k+1) + \{D'\}$

Step 4:

After Steps 2-3 have been completed for each $D \in \text{GRAPHS}_k$,
check if $\text{GRAPHS}(k+1) = \emptyset$.

If $\text{GRAPHS}(k+1) = \emptyset$, then terminate.

If $\text{GRAPHS}(k+1) \neq \emptyset$, then continue working through
Steps 2-3 for each $D \in \text{GRAPHS}_{k+1}$

Step 5:

Output the automata as nodes & edges (w/ the ltt structure
& map info included)
But what we really want is the strongly connected components